

Homework — Stage 08: Exploratory Data Analysis (EDA)

All paths below are inside `homework/homework08/`.

Assignment

Create a comprehensive **EDA notebook** for your project dataset.

Requirements

- Statistical summaries: `.info()`, `.describe()`, missing counts
- Categorical profile: value counts for every non-numeric column
- Distributional plots: histograms/boxplots for key variables
- Bivariate visuals: scatter/line for at least 2 relationships
- At least one variable plotted **over time**, with a note on trend, seasonality or level shifts
- `src/eda.py` holding a reusable `eda_summary(df)`, imported into your notebook
- (Optional) Correlation heatmap
- Clear markdown documenting **findings, risks, and assumptions**

Step-by-Step

1. Copy `stage08_exploratory-data-analysis_homework-starter.ipynb` into `homework/homework08/`, rename it `homework08_exploratory-data-analysis_submission.ipynb`, and open it.
2. Load your dataset (or adapt the starter's synthetic generator to mimic your schema).
3. Profile numeric columns (`.info()`, `.describe()`, missing counts, skew/kurtosis) **and** categorical columns (value counts).
4. Plot at least 3 distributions and 2 relationships.
5. Plot at least one variable **over time** and say what you see: trend, seasonality, a level shift, or nothing.
6. Extract the lecture's `eda_summary(df)` helper into `src/eda.py`, import it, and call it on your dataframe.
7. Note any skew, outliers, seasonality, or structure.
8. Conclude with **"Implications for next step"**: what you'll engineer or clean.

Grading Rubric (100 pts)

- (25) Completeness of profiling — numeric **and** categorical
- (25) Quality and clarity of visuals
- (15) Time-series read: the plot, plus what it implies for stages 09 and 10b
- (25) Insightfulness of interpretation (not just "what," but "so what")
- (10) Organization and readability, and `src/eda.py` imports cleanly

Stretch Goals

- Add a correlation heatmap and say which pairs actually matter for modeling
- Profile the same dataset at two granularities (daily vs monthly) and compare what each reveals

- Extend `eda_summary()` to flag columns needing attention before stage 09 — high missingness, near-zero variance, or one category dominating

Example Submission Expectations

- A Jupyter notebook named `homework08_exploratory-data-analysis_submission.ipynb`
- Clean titles/labels; brief markdown per plot
- Final section: “Top 3 insights” + “Assumptions & risks”

Chain Statement

In the lecture, we learned how to profile and visualize data to surface patterns. Now, you will adapt those patterns to your own dataset and document what they imply for preprocessing and feature engineering.